

Lab 8 : CI/CD Pipeline

I. Create a New GitHub Repository

1. Go to GitHub and log into your account.
2. Create and name the repository (e.g., my-github-actions-project).
3. Clone the Repository Locally

git clone <https://github.com/yourusername/my-github-actions-project.git>

4. Initialize Node.js Project

yarn init -y # or npm init -y

yarn add express # or npm install express

5. Create a file **app.js** in the root directory with a simple Node.js server.

```
const express = require('express');
const app = express();
const port = 3000;
app.get('/', (req, res) => {
  res.send('Hello, GitHub Actions!');
});
app.listen(port, () => {
  console.log(`App listening at 

6. Add basic test

```

yarn add mocha chai # or npm install mocha chai

7. Create a **test.js** file in the test folder.

```
const assert = require('chai').assert;
const app = require('../app');

describe('Simple test', function() {
  it('should return hello message', function() {
    assert.equal('Hello, GitHub Actions!', 'Hello, GitHub Actions!');
  });
});
```

```
});
```

8. Commit and Push the Code

```
git add .  
git commit -m "Initial project setup with app.js and basic test"  
git push origin main
```

II. Set up GitHub Actions Workflow

1. In the root of your repository, create a directory structure like **.github/workflows** to store your workflow configuration.

```
mkdir -p .github/workflows
```

2. Inside the workflows folder, create a file called **ci.yml**

```
touch .github/workflows/ci.yml
```

3. Open **ci.yml** in your preferred editor, and add the following YAML configuration:

```
name: Node.js CI
```

```
on:
```

```
  push:
```

```
    branches:
```

```
      - main # Trigger this workflow only for the main branch
```

```
  pull_request:
```

```
    branches:
```

```
      - main # Trigger this workflow for PRs targeting the main branch
```

```
jobs:
```

```
  build:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - name: Checkout code
```

```
        uses: actions/checkout@v3
```

```
      - name: Set up Node.js
```

```
        uses: actions/setup-node@v3
```

```
        with:
```

node-version: '16' # Specify Node.js version

- name: Install dependencies

run: |

yarn install # or npm install

- name: Run tests

run: |

yarn test # or npm test

4. Commit and push the Workflow Configuration:

git add .github/

git commit -m "Add GitHub Actions workflow for Node.js CI"

git push origin main

III. Monitor Your Workflow in GitHub Actions

1. Once you've pushed the changes, GitHub Actions will automatically trigger the workflow because it listens for changes in the main branch.
2. View Workflow Runs:
 - a. Go to your GitHub repository in the browser.
 - b. Click on the **Actions** tab at the top of the repository.
 - c. You should see the new workflow running. Click on the workflow to see logs and check the status of each step.

IV. Test the CI Workflow

1. Make Code Changes, commit and push your changes to trigger a new workflow run
- 2. After pushing, visit the Actions tab again to see the new workflow run.**

Submission:

1. Screenshot important steps
2. Explain step by step of command in ci.yml
3. Explain in your own words what have you learned from this lab.